Conditions for Factor Analysis Transcript

Hello everyone, welcome to lesson 1 video 2, in this video we will discuss the conditions for applying factor analysis. Factor analysis should not be used blindly, there are certain assumptions and requirements that needs to be checked first. Generally before doing factor analysis we try to find out whether the data that we have is actually suitable for doing factor analysis or not.

So whether the key assumptions that we should lay down or overlay they are being met or not. So in this entire video we are going to learn about conditions which affect factor analysis or conditions or assumptions which should be obeyed before we start doing factor analysis. So this is the core concept you all already know that if your input is all vague, output is going to be vague, garbage in garbage out.

So before running factor analysis we must assess the factor ability whether we can actually create factors in the data or not. If factors are not even, we cannot form even factors then there is no point in using factor analysis on that data. So that's the important, that's the most important thing within it.

Now these are the assumptions that we lay. The very first thing is level of measurement, variable should be continuous, we should not have categorical variable, we actually can have categorical variables but we need to convert them into numeric representation and there are various different ways for them. We have let's say the second thing is sample size, the correlation matrix needs to be stable and we need minimum amount of sample for the correlation matrix to be established because if the input features or input variables are not correlated then factor analysis cannot sustain there.

Let's say if there is no commonalities between the input features then we cannot even do factor analysis. So we need to have enough sample size so that we can create a correlation matrix first and the minimum possible data that we need for doing that is at least 5 to 10 observations per variable is required. And this is not hardcore limit but for making the correlation matrix stable or at least getting some realistic estimate of the correlation matrix we should have at least 5 to 10 observations per variable.

Then we have sufficient correlation for factors to exist the variables must be correlated with each other this is what we have already discussed. Linearity the model assumes that there is linear relationship between the observed variables if there is huge non-linearity then this can distort the results of the entire factor analysis. So these are few checklist or few assumptions that we need to look into the very first was level of measurement you should be having continuous variable in your data.

If you do not have continuous variables if you have categorical variables you should first change them to continuous there are methods for doing that or else you can remove features which are not continuous. Then sample size you should have minimal possible sample size so as to estimate the correlation matrix correctly if you do not have that the results could be really incorrect. Then you have sufficient condition which means that for the factors to exist the correlation matrix should have certain high correlation

So let's say for example if your data doesn't has high correlation it means that no two features are having similar information or giving a similar information right. In that case you cannot have factors created or you cannot have distinct factors created in that case you will have as many factors as the variables that you have. Then linearity this is an assumption that the observed variables that we have in our data they are linear if they are not linear then our output or the output of the factor analysis could be misleading.

Now these are the two different types of statistical tests that we do for checking whether we have the information enough in the data to create factor analysis or do factor analysis. The very first thing is we have Barlet's test and the second thing is KMO or Kaiser Meyer Olkin test. These two tests test two different things.

One is testing for the variables are correlated or not. The second is testing for variance in the features. So both of them are two different types of tests all together.

So let's first study about Barlet's. Barlet's test is testing for whether the features are correlated or not.

So let's say for example there is correlation in the data or not. For doing that it checks whether the correlation matrix is actually a identity matrix or not. OK. Now if the correlation matrix is an identity matrix which means that every element in the diagonal is 1 and all the other non-diagonal elements are 0. It means that there is no correlation between two different features because in correlation matrix you will have off diagonal elements.

You should have I mean off-diagonal elements greater than 0 or at least less than 0 and greater than minus 1. So in that case if that is not existent and everything is 0 it means that no correlation exists in the data and hence you cannot use factor analysis. So Barlet's test only tests for whether correlation exists in the data or not. OK.

For doing that it compares your output correlation matrix with an identity matrix and then it comes up with a statistically significant output. If the p-value is less than 0.05 you say that there is some correlation between variables. If the p-value is greater than 0.05 you say that the input matrix is somewhat related to or it's looking like an identity matrix.

Hence there is no correlation. Now the second test that we use is Kaiser Meier. In this test we kind of study the variance, the measure of proportion of variance among variables that might be common to different types of variables.

So we want a very high estimate in this particular case. Let's say if the estimate is lesser than 0.05 or lesser than 0.5 in that case we say that values below 0.5 mean the features are not correlated or they do not we actually cannot create factors out of it because if there is no variance. OK then there is no point in capturing those variance via factors.

So this entire test is limited to testing for variance while the first test is limited to testing for whether the matrix input matrix which is correlation matrix is actually having correlated data in it or not. If there is correlated data then the matrix element non-diagonal elements would be non-zero otherwise if they would be zero if they are zero then we have an identity matrix and p value would be greater than 0.05. While in this particular case we are testing proportion of variance among variables. So once we have that and we see that the proportion of variance is significant enough then we say or in those cases KMO value would be greater than 0.6 if that comes out to be greater than let's say we are testing out on a data set and our KMO value turns out to be greater than 0.6. In all those cases we will say that there are variances which exist ok hence we can have factors in the data.

I hope this is clear to everyone. In summary before applying factor analysis we must ensure that the data set is suitable checking sample sizes correlations statistical test helps us to understand whether analysis is appropriate or not. In this next video we will compare factor analysis with principal component analysis.

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